

MRIDUL KHURANA

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EDUCATION

Virginia Tech

Aug 2022 – May 2027

PhD. in Computer Science, *Advisor: Dr. Anuj Karpatne* GPA 4.0/4.0

Blacksburg, Virginia

Courses: Computer Vision (CV), Natural Language Processing (NLP), Deep Reinforcement Learning (RL), Adv Machine Learning

Graduate Teaching Assistant: Natural Language Processing, Social Media Analytics

Delhi Technological University

Aug 2015 - May 2019

Bachelor of Technology (B.Tech.), Electrical Engineering

New Delhi, India

RESEARCH INTERESTS

My research focuses on multimodal image and video diffusion models for scientific domains, using personalized & structure aware conditioning with taxonomies, environmental data, & DNA barcodes to enable fine-grained & long-tail generation.

PROFESSIONAL EXPERIENCE

Advanced Micro Devices (AMD) Inc

San Jose, CA

Research Intern

Jan 2025 – May 2025

- **Model optimization** using sparsity-aware **quantization (SpQR)** of LLMs like **Llama-3**, **Qwen**, etc for near-lossless inference.
- Extended **AMD Quark** quantization capabilities to **Mixed-Precision** using SpQR and improving over **GPTQ** using **ROCm**.

Knowledge-guided ML and CV Lab, Virginia Tech

Blacksburg, VA

Graduate Research Assistant | Advisor: Dr. Anuj Karpatne

Aug 2022 – Present

- Developing wildlife **video diffusion** models with controllable motion and species-level fidelity on **text-to-video** backbones.
- Integrating **Sparse Autoencoders (SAE)** for interpretable **diffusion** models with Representation Autoencoders (**DiT w/ RAE**).
- Progressive training of **Text-to-Image models** for **Fine-Grained generation** for **long-tailed** distributions. (**ICCV'25**)
- **Hierarchical** conditioning of **Stable Diffusion** with **graphs** for accelerating scientific discoveries (**ECCV'24**).
- Collaborated in designing an **automated data processing pipeline** using large visual and multimodal models like **Grounding DINO** and Segment Anything (**SAM**) for an **attribute identification** and **visual segmentation** dataset (**CVPR'25**)
- **Multimodal scientific benchmark dataset** for Vision Language Models (VLMs) including **GPT-4o**, **LLaVa**, **BLIP** (**NeurIPS'24**)

Theremin.ai (Quant startup)

Mumbai, India

Software Engineer

May 2020 - July 2022

- Designed and built a **Trading Platform (OMS)** handling over **\$10 Million** in Indian stock markets. Mentoring **2 interns**.
- Achieved a **15x increase** in data volume to support **1,500 securities**, deploying **event-driven trading** framework on **AWS**.

Fractal Analytics

Mumbai, India

Software Engineer

June 2019 – Apr 2020

- Built ML models with **10,000+ variables** reducing dimensionality by **97%** using **PCA**, variable binning & **clustering**.
- Used **Random Forest** and **CNNs**, achieving **10% increase** on an F1 score from previous models.

Vision and AI Lab (VAL), IISc Bangalore

Bangalore, India

Research Intern | Advisor: Dr. R. Venkatesh Babu

May 2018 – July 2018

- Developed **monocular depth estimation** models using **CNNs** (ResNet-50) & fovea transformations for outdoor scenes.
- The absolute & squared relative errors were reduced by 0.7% and 2% resp. on the KITTI stereo dataset (at par with SOTA).
- Employed Fully Convolutional Residual Networks (**FCRN**) using **unsupervised learning** for indoor scenes in Tensorflow.

PROJECTS

Mitigating Simple and Adversarial Attacks on Vision Language Models (VLMs)

[GitHub](#)

Feb 2024 - May 2024

- Proposed **Visual Latent Smoothing** enhancing robustness of LLaVa & VLMs against **adversarial** attacks by **9% & 4%** on MCQA.

Numeral-Aware Large Language Models (LLMs)

[GitHub](#)

Sep 2023 – Dec 2023

- **Prompt tuning** and Instruction fine-tuning **Llama-2** using **LoRA** for accurately generating numeral-aware news headlines.

INVITED TALKS AND AWARDS

- **Tree-of-Life Meets AI:** Knowledge-guided Generative Models, hosted by Voxel51, Nov. 2024
- Received **Travel Award** from CS Department, Virginia Tech for attending ECCV 2024.
- **Taekwondo:** State level **Gold medalist**, India. **Music:** Certified **Grade 2** drummer, Rockschoool, **Trinity College of London**.

PROFESSIONAL SERVICE

- **Reviewer** for ICLR 2026, 2025; NeurIPS 2025, 2024, 2023 Workshops ; ICML 2025 Workshops; AAAI 2024-2025 Workshops.
- **Program Committee Member:** Imageomics Workshop at NeurIPS 2025, AAAI 2025 and AAAI 2024.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Shell Scripting, SQL, NoSQL, Git, MATLAB
- **Technologies:** PyTorch, HuggingFace, vLLM, Tensorflow, AWS, Docker, OpenCV, Scikit-learn, WandB, MongoDB

PUBLICATIONS**Conferences**

7. **Mridul Khurana**, Amin Karimi Monsefi, Justin Lee, David Carlyn, Julia Chae, Jianyang Gu, Rajiv Ramnath, Sara Beery, Wei-Lun Chao, Anuj Karpatne, Cheng Zhang. “TaxaAdapter: Coupling Vision Taxonomy Models with Diffusion for Fine-Grained Species Generation.” **(under review - CVPR 2026)**
6. **Mridul Khurana**, Arka Daw, M. Maruf, Josef C Uyeda, Wasila Dahdul, ..., Anuj Karpatne. “Hierarchical Conditioning of Diffusion Models Using Tree-of-Life for Studying Species Evolution.” **ECCV 2024 (Paper)**
5. Amin Karimi Monsefi, **Mridul Khurana**, Rajiv Ramnath, Anuj Karpatne, Wei-Lun Chao, Cheng Zhang. “TaxaDiffusion: Progressively Trained Diffusion Model for Fine-Grained Species Generation.” **ICCV 2025 (Paper)**
4. Medha Sawhney, Abhilash Neog, **Mridul Khurana**, Anuj Karpatne “Beyond Loss Guidance: Using PDE Residuals as Spectral Attention in Diffusion Neural.” **(under review - ICLR 2026) (Paper)**
3. Kazi S. Mehrab, M. Maruf, Arka Daw, Harish. B. Manogaran, Abhilash Neog, **Mridul Khurana**, ..., Anuj Karpatne “Fish-Vista: A Multi-Purpose Dataset for Understanding & Identification of Traits from Images.” **CVPR 2025 (Paper)**
2. M. Maruf*, Arka Daw*, Kazi S. Mehrab, Harish. B. Manogaran, Abhilash Neog, Medha Sawhney, **Mridul Khurana**, ..., Anuj Karpatne. “VLM4Bio: A Benchmark Dataset to Evaluate Pretrained Vision-Language Models for Trait Discovery from Biological Images.” **NeurIPS 2024 (Paper)**
1. Mohannad Elhamod, **Mridul Khurana**, Harish Babu Manogaran, Josef C Uyeda, ..., Anuj Karpatne. “Discovering Novel Biological Traits from Images using Phylogeny-guided Neural Networks.” **KDD 2023 (Oral) (Paper)**

Journals

* Denotes equal contribution

1. **Mridul Khurana***, S M Rayeed*, Alyson East*, Isadora E Flux, Elizabeth G Campolongo, Samuel Stevens, ..., Anuj Karpatne, Charles V Stewart, Graham Taylor, Sydne Record. “A continental-scale dataset of ground beetles with high-resolution images and validated morphological trait measurements.” **NATURE Scientific Data (under review)**

Workshops

5. Medha Sawhney, Abhilash Neog, **Mridul Khurana**, Arka Daw, Anuj Karpatne “Investigating PDE Residual Attentions in Frequency Space for Diffusion Neural Operators.” **NeurIPS 2025 Workshop** on ML4PS.
5. Amin Karimi Monsefi, **Mridul Khurana**, Rajiv Ramnath, Anuj Karpatne, Wei-Lun Chao, Cheng Zhang. “TaxaDiffusion: Progressively Trained Diffusion Model for Fine-Grained Generation.” **CVPR 2025 Workshop (Oral)** on CV4Animals **(Paper)**
4. Medha Sawhney, Abhilash Neog, **Mridul Khurana**, Amartya Dutta, Arka Daw, Anuj Karpatne. “Physics-guided Diffusion Neural Operators for Solving Forward and Inverse PDEs.” **CVPR 2025 Workshop** on CV4Science.
3. Amartya Dutta, K.S. Mehrab, Medha Sawhney, Abhilash Neog, **Mridul Khurana**, ..., Anuj Karpatne. “Open World Scene Graph Generation using Vision Language Models.” **ICML 2025 Workshop** on World Models. **(Paper)**
2. **Mridul Khurana**, Arka Daw, M. Maruf, ..., Anuj Karpatne. “Conditioning Diffusion Models Using the Knowledge of Phylogeny for Understanding Species Evolution”. **AAAI 2024 Workshop (Oral)** on Imageomics **(Poster)**
1. Mohannad Elhamod, **Mridul Khurana**, Harish Babu Manogaran, ..., Anuj Karpatne. “Discovering Novel Biological Traits from Images using Phylogeny-guided Neural Networks.” **CVPR 2023 Workshop (Oral)** on CV4Animals **(Poster)**